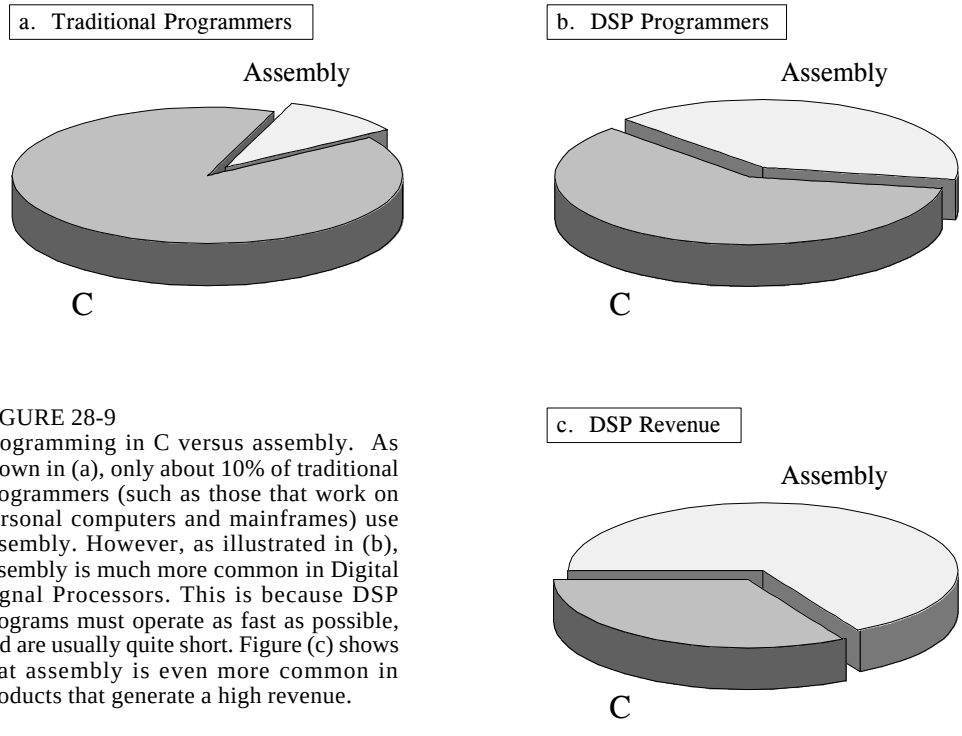

FIGURE 28-9
Programming in C versus assembly. As shown in (a), only about 10% of traditional programmers (such as those that work on personal computers and mainframes) use assembly. However, as illustrated in (b), assembly is much more common in Digital Signal Processors. This is because DSP programs must operate as fast as possible, and are usually quite short. Figure (c) shows that assembly is even more common in products that generate a high revenue.

array by the corresponding coefficient in the other array, and sum the products, i.e. $x[0]\times y[0] + x[1]\times y[1] + x[2]\times y[2] + \cdots$. This should look very familiar; it is the fundamental operation in an FIR filter. That is, each sample in the output signal is found by multiplying stored samples from the input signal (in one array) by the filter coefficients (in the other array), and summing the products.

Table 28-2 shows how the dot product is calculated in a C program. In lines 001-004 we define the two arrays, *x*[ ] and *y*[ ], to be 20 elements long. We also define *result*, the variable that holds the calculated dot

TABLE 28-2
Dot product in C. This progam calculates the dot product of two arrays, x[ ] and y[ ], and stores the result in the variable, *result*.

```
001    #define LEN 20
002      float dm x[LEN];
003      float pm y[LEN];
004      float result;
005
006    main()
007
008    {
009      int n;
010      float s;
011      for (n=0;n<LEN;n++)
012        s += x[n]*y[n];
013      result = s
014    }
```